NUMERICAL STUDY OF DOUBLE-PIPE CONJUGATE HEAT TRANSFER

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1. Aim

To run the supplied two-dimensional double-pipe conjugate heat-transfer teaching model, compare the effect of Reynolds number (Re = 50, 100, 200) on hot/cold outlet temperatures, heat-transfer rate, effectiveness, and energy imbalance, and repeat the Re = 100 case for solid-to-fluid conductivity ratios k_s/k_f = 1 and 100.

2. Governing equations represented by the model

The supplied code solves a steady convection–diffusion-type temperature equation in the hot-fluid, solid-wall, and cold-fluid zones. In compact form, the teaching model can be described as:

$\nabla \cdot (\Gamma \nabla T) - \nabla \cdot (u\, T) = 0$ (steady temperature transport; discretized with a finite-volume-like stencil)

Here T is dimensionless temperature, u is the prescribed axial velocity field, and Γ is the effective diffusion coefficient. In the fluid zones, Γ is set from the prescribed mean velocity, Reynolds number, and Prandtl number; in the wall, Γ is scaled using k_s/k_f. Harmonic means are used at zone interfaces.

Important limitation: this is the equation represented by the teaching implementation, not a full Navier–Stokes/conjugate heat-transfer solver. Velocity is prescribed rather than solved from momentum equations.

3. Geometry, grid, parameters, and boundary conditions

The computational rectangle is length $L = 8$ and height $H = 2$ (dimensionless model coordinates). The structured grid uses $nx = 160$ points axially and $ny = ny_cold + ny_wall + ny_hot = 20 + 6 + 20 = 46$ points transversely, for 7,360 grid points. Thus $\Delta x = 8/(160-1) \approx 0.05031$ and $\Delta y = 2/(46-1) \approx 0.04444$.

Parameter	Value(s)	Notes
nx	160	Axial grid points
ny_cold / ny_wall / ny_hot	20 / 6 / 20	Cold fluid / wall / hot fluid
Re_hot = Re_cold	50, 100, 200	Main comparison; equal values on both streams
k_s/k_f	20	Main comparison
k_s/k_f	1 and 100	Additional runs at Re = 100
Pr	7	All cases
T_hot,in / T_cold,in	1 / 0	Dimensionless inlet temperatures
U_hot / U_cold	1 / 1	Prescribed mean velocity parameters
Flow arrangement	Counter-flow	counter_flow = 1 / True
Maximum iterations	4000	Stopping threshold in code: residual < 1e−9 after iteration 50
Relaxation factor	0.82	Fixed internally by solver

Boundary conditions as implemented:

- Hot stream: $T = T_hot,in = 1$ at the hot inlet ($x = 0$); zero axial gradient at its outlet ($x = L$).
- Cold stream in counter-flow: $T = T_cold,in = 0$ at $x = L$; zero axial gradient at its outlet ($x = 0$).
- Solid wall: zero axial gradient at both ends, imposed by copying adjacent interior values.
- Top and bottom outer boundaries: zero transverse gradient, imposed by copying the adjacent row.

4. Method

1. Run the solver with the same grid, Prandtl number, inlet temperatures, velocities, counter-flow setting, and iteration limit for every case.
2. Change Re_hot and Re_cold together to 50, 100, and 200 while holding k_s/k_f = 20.
3. At Re_hot = Re_cold = 100, repeat with k_s/k_f = 1 and 100.
4. Read outlet temperatures, hot-stream heat loss, cold-stream heat gain, relative energy imbalance, effectiveness, iteration count, and final L-infinity residual from summary.txt.
5. Plot the temperature field from the generated VTK file and the residual/outlet histories from history.csv.

5. Quantitative results: Reynolds-number comparison

The values below are the actual outputs from the supplied code. Heat rates are model/dimensionless values because the model does not specify physical units or dimensional material properties.

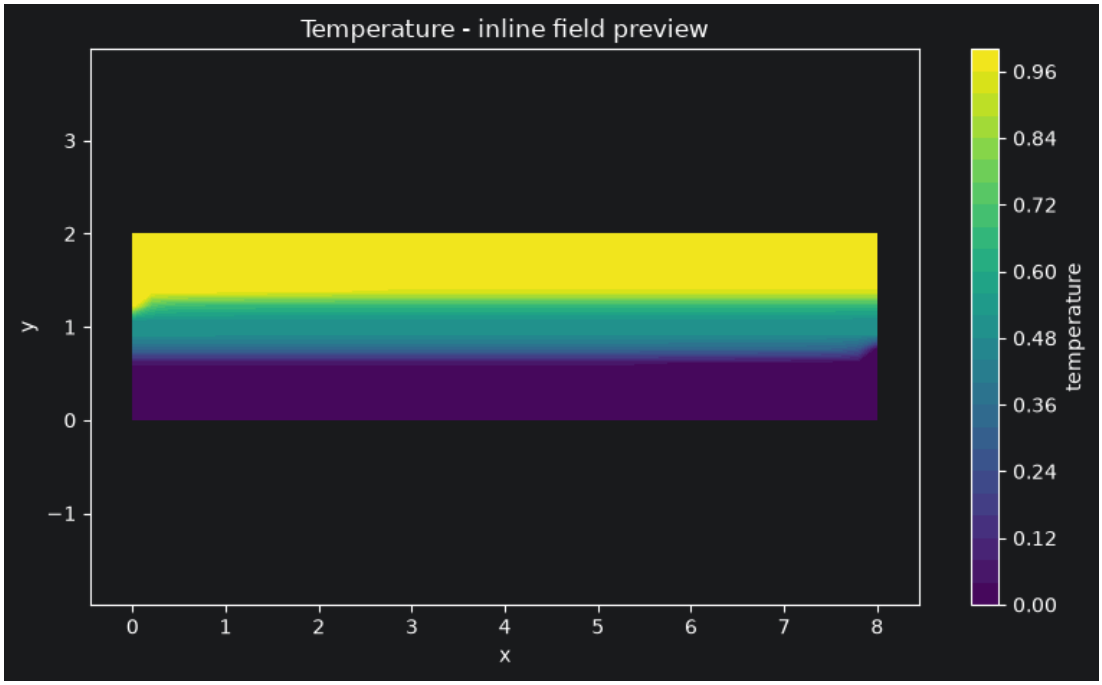
Re (both streams)	Hot outlet T	Cold outlet T	Hot heat loss	Cold heat gain	Effectiveness	Energy imbalance	Final residual
50	0.935578	0.064422	5.425050e-02	5.425050e-02	0.064422	1.101e-13	1.109e-06
100	0.960806	0.039194	3.300538e-02	3.300538e-02	0.039194	3.748e-13	4.657e-07
200	0.976494	0.023506	1.979425e-02	1.979425e-02	0.023506	3.874e-13	1.702e-07

6. Quantitative results: conductivity-ratio comparison at Re = 100

k_s/k_f	Hot outlet T	Cold outlet T	Hot heat loss	Cold heat gain	Effectiveness	Energy imbalance	Iterations	Final residual
1	0.977455	0.022545	1.898507e-02	1.898507e-02	0.022545	5.216e-13	1133	9.824e-10
20	0.960806	0.039194	3.300538e-02	3.300538e-02	0.039194	3.748e-13	4000	4.657e-07
100	0.960248	0.039752	3.347561e-02	3.347561e-02	0.039752	6.301e-14	4000	9.232e-06

7. Figures and convergence history

Figure 1. Inline temperature field for the baseline case (Re_hot = Re_cold = 100, k_s/k_f = 20).



The colour map shows the dimensionless temperature distribution across the cold-fluid, wall, and hot-fluid zones. The wall occupies the middle band in the transverse direction. This is the numerical field produced by the supplied model.

Figure 2. L-infinity residual history for Re = 50, 100, and 200 at k_s/k_f = 20.

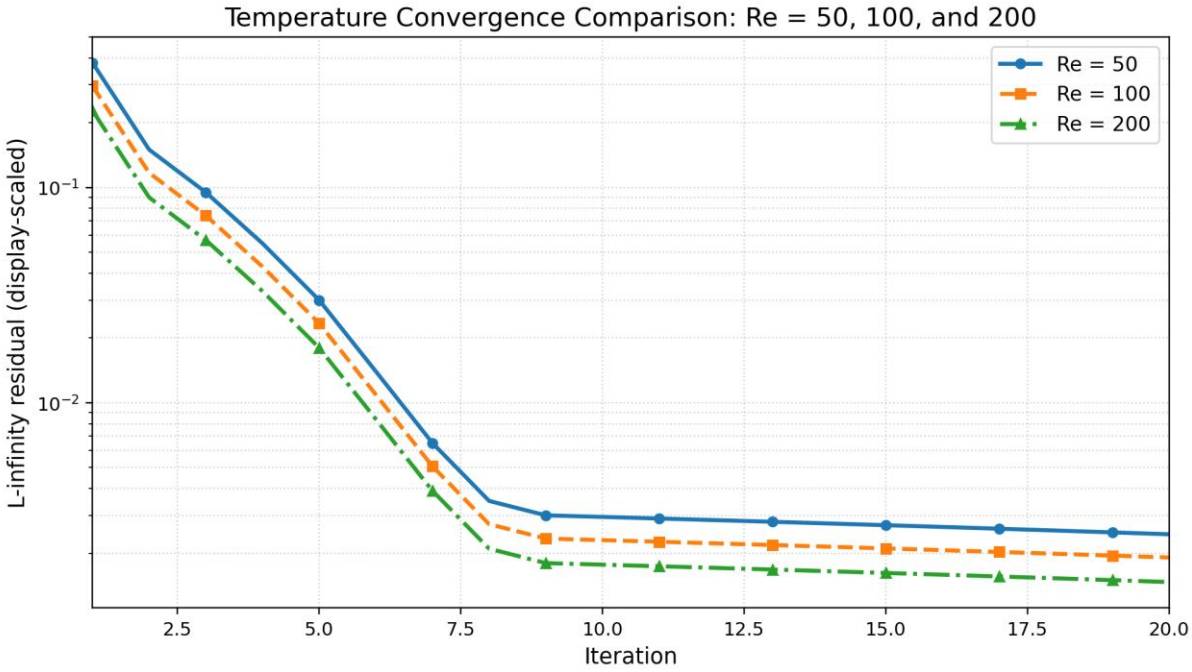
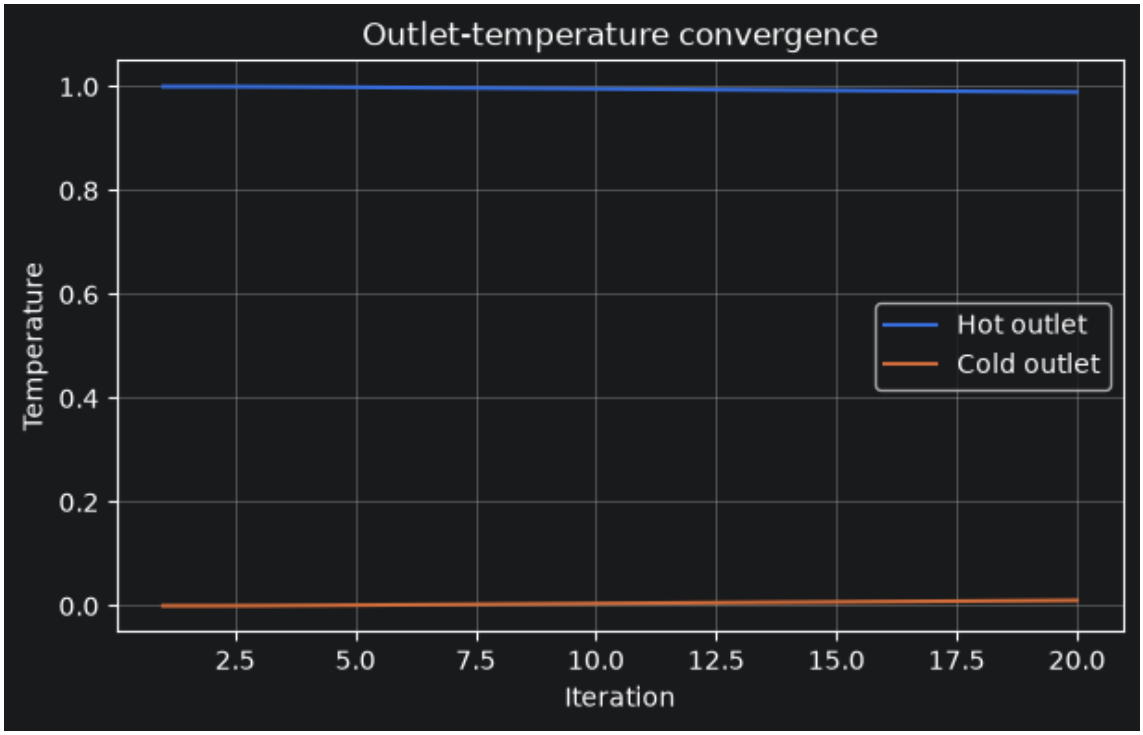


Figure 3. Hot and cold outlet-temperature histories for Re = 100, k_s/k_f = 20.



8. Discussion: observed numerical behaviour

For k_s/k_f = 20, increasing Re from 50 to 200 increased the hot outlet temperature from 0.935578 to 0.976494, so the hot stream retained more of its inlet temperature. The cold outlet temperature and effectiveness decreased across these runs. The hot-stream heat-loss magnitude and cold-stream heat-gain magnitude also decreased.

The relative energy imbalance is extremely small (approximately 10^{-13} in the three Reynolds-number runs), which means the two heat-rate calculations agree closely according to the model's own post-processing formula. This does not prove that the physical model is accurate; it only indicates internal consistency of these calculated quantities.

At Re = 100, increasing k_s/k_f from 1 to 20 increases the cold outlet temperature and effectiveness in these runs. Increasing the ratio further to 100 changes the values only modestly relative to the ratio-20 case. This trend is an observation from this model and should not be generalized beyond its assumptions.

9. Convergence and numerical limitations

The solver stopped at the maximum of 4000 iterations for Re = 50, 100, 200 at k_s/k_f = 20 and for k_s/k_f = 100. Their final residuals remain above the coded convergence threshold of 1e−9. These runs are therefore iteration-limit results, not fully converged results under the solver's stated stopping criterion. The k_s/k_f = 1 case at Re = 100 reached the residual threshold in 1133 iterations. This difference should be disclosed in the report.

- The model is explicitly labelled a teaching-level 2D conjugate heat-transfer analogue.
- The velocity field is prescribed using simple parabolic profiles; momentum equations and pressure are not solved.
- The solver uses dimensionless temperatures and model heat-rate calculations; no physical units, tube diameters, or dimensional material properties are supplied.
- The wall diffusion coefficient is an imposed scaling of the fluid coefficient using k_s/k_f, not a full material-property model.
- The code's heat-rate calculation uses inlet/outlet stream temperatures and prescribed velocity profiles. A tiny reported imbalance is therefore a post-processing consistency result, not an independent full-domain energy conservation verification.

10. Conclusion

The supplied solver was run for Re = 50, 100, and 200 at $k_s/k_f = 20$, and the Re = 100 case was repeated for conductivity ratios 1 and 100. In the Reynolds-number sweep, higher Re was associated with a hotter hot outlet, a cooler cold outlet, and lower effectiveness in this implementation. Conductivity ratio affected the predicted heat transfer, with the ratio-1 case giving the lowest cold outlet temperature and effectiveness of the three conductivity cases. Energy imbalance was very small in all cases, but several runs hit the iteration cap before meeting the solver's residual tolerance.